GCE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2022

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2022 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

Contents

Assessment Unit AS 1	Approaches to Systems Development	3
Assessment Unit AS 2	Fundamentals of Digital Technology	4
Assessment Unit A2 1	Information Systems	6
Assessment Unit A2 2	Application Development	9
Contact details		11

GCE Digital Technology

Chief Examiner's Report

Subject Overview

Due to the extremely difficult circumstances faced by teachers and candidates as they prepared for this set of examinations, direct comparisons of the performance of this cohort with those of pre-Covid candidates is not possible, particularly taking into account the range of special arrangements CCEA put in place to try to mitigate the impact of these circumstances. Instead, this report highlights the strengths and weaknesses of the 2022 cohort while avoiding any speculation about the impact of Covid on the candidates.

However, it is important to repeat the main general points made in previous reports.

- Candidates must be familiar with DT terminology and must use appropriate DT terminology in their responses.
- Where a question is framed around a context, candidates must address this context as this is an important element of the specification's AO2 assessment.
- Questions which are identified as assessing QWC are also a key element of the specification's AO3 assessment. The required standard of response for these questions is much higher as is indicated by the unit comments below.

Assessment Unit AS 1 Approaches to System Development

Unit Overview

As this was an optional unit, the number of candidates taking the unit was very small making an overall comparative assessment of both the candidature and the question paper quite meaningless. Instead, some brief comments about each question are provided.

Q1 **(a)** Candidates could explain a constraint and provide a suitable example.

(b) Most candidates could identify at least two appropriate risks.

(c) Candidates were appropriately familiar with Gantt charts.

(d) Most candidates could complete the Activity Dependency Table accurately.

Q2 **(a)** Candidates could not distinguish between data modelling and process modelling.

(b) Candidates were not familiar with the terms build and time frame.

(c) Most candidates were familiar with version management in the context of systems development.

(d) This QWC question required candidates to compare the waterfall model and the agile approach with respect to changing user requirements, in a concise, focused answer using the relevant DT terminology. Only a few candidates met the strict criteria for a middle-band response.

Q3 **(a)** Most candidates were familiar with the term user interface.

(b) Most candidates could describe how storyboarding is used in the design of the user interface.

(c) Most candidates could describe how prototyping is used in the design of the user interface.

(d) Some complete and correct DFDs were produced.

(e) Most candidates could describe the parallel changeover method.

Q4 **(a)** Most candidates had difficulty describing the three terms accurately.

(b) Most candidates were familiar with the two terms.

(c) This QWC question required candidates to compare the use of a count-controlled loop and a condition-controlled loop, in a concise, focused answer using the relevant DT terminology. Only a few candidates met the strict criteria for a middle-band response.

(d) Very few candidates could describe the purpose and output of the algorithm.

(e) Only a few candidates could explain the effect of moving the output statement.

Q5 **(a)** Most candidates were familiar with the two types of translator but few could provide the required detail.

(b) Only a few candidates could explain clearly why the client would not be provided with the source code.

(c) Most candidates struggled with the OO terminology involved.

Assessment Unit AS 2 Fundamentals of Digital Technology

Unit Overview

These comments reflect the opening comments about the extremely difficult circumstances faced by teachers and candidates as they prepared for this set of examinations.

Q1 **(a)** An excellent overall standard of response. Most candidates were familiar with the three terms and could convert them accurately to the required units.

(b) An excellent overall standard of response. Most candidates could suggest an appropriate use for the eighth bit, with error checking being the most common response.

(c) Although the majority of candidates could identify an advantage and/or a disadvantage, some failed to provide the required detail. Answers such as 'Represent more characters', for example, could not be awarded [2].

(d) The responses to this question demonstrated that candidates were in general able to do the appropriate conversions from binary to decimal. However, candidates must be aware that in a sign and magnitude representation, the sign of the number as well as its magnitude must always be specified. Many candidates lost a mark for writing 19 instead of +19. This issue has been clearly identified in previous Examiner's Reports.

(e) The majority of candidates were able to perform the required decimal to binary calculation and most of these showed all stages in their working as required.

Q2 **(a)** Most candidates could name at least three factors affecting the quality of information. A small number of candidates failed to read the wording of the question carefully and suggested presentation which was excluded by the question's stem.

(b) The overall level of response was of high quality and many excellent descriptions of both proof reading and double entry were noted. However, it must be stressed that the key point about proof reading is that the input is compared against the source document. Responses such as 'checked over' are not appropriate at this level. With double entry, the key point is that the comparing of the two versions of the input is an automatic process and a phrase equivalent to 'compared by the computer' is required.

(c) Of the three methods of validation, range and format were described most accurately. Candidates must be reminded that 'between two values' is not appropriate at this level for range, and that giving an example for any of the checks is not sufficient for an 'explain' question. In spite of feedback in previous Chief Examiner Reports, a significant proportion of candidates incorrectly described the use of a drop-down list, or similar, for lookup.

(d) The majority of candidates demonstrated appropriate competence in calculating a check digit and could show all the stages as required. A significant proportion of candidates lost a mark by failing to make a final statement about the validity of the given product code, leaving the marker to deduce this from their answer.

Q3 **(a)** The majority of candidates found it difficult to describe the use of the address bus, even the fundamental point of 'the address of what'. ROM was much better understood, but the long-standing problem of candidates completely confusing ROM with RAM remains. The use of a port was better understood.

(b) Cache memory was well described by the majority of candidates. The main weakness was the confusion by a significant number of candidates of the processor's use of cache memory with the use of a cache by applications such as a web browser. This issue has been highlighted before.

(c) The overall level of response to this question was very disappointing. Most candidates struggled to relate either register to the fetch-execute cycle as required. A worrying proportion of candidates produced responses such as 'the program counter keeps a check on…'.

(d) While most of candidates were familiar with secondary storage, many struggled to relate their responses to a smartphone as was required.

(e) This QWC question required candidates to evaluate the use of flash storage by a smartphone in a concise, focused answer using the relevant DT terminology. As was expected, only a very small number of candidates met the very strict criteria for a top-band response. Also, as was intended, only a few more met the strict criteria for a middle-band response.

Q4 **(a)** Of the two types of operating system, only multitasking was understood by the candidature as a whole and some excellent and complete responses were noted. A common weakness, however, was a failure to set multitasking clearly in the context of a single user. Multithreading was poorly understood and few candidates gained any marks for their response.

(b) A significant proportion of candidates confused archiving with the backing up of data but these candidates sometimes picked up one or two incidental marks. Of those candidates properly familiar with archiving, most could accurately describe its use by a typical organization.

(c) The majority of candidates could identify an advantage and a disadvantage of data compression, but many of these failed to address the issue of an image inserted in a web page.

(d) The overall level of response to this question was disappointing as many candidates struggled to produce a clear and concise explanation of application software.

(e) The majority of candidates focused their responses on the input of spoken commands and ignored the broader considerations of voice recognition.

(f) This QWC question required candidates to evaluate the use of a WIMP interface and a CLI for an application intended to be used by primary school children. The question required a concise, focused answer using the relevant DT terminology. As was expected, only a very small number of candidates met the very strict criteria for a top-band response. Also, as was intended, only a few more met the strict criteria for a middle-band response.

Q5 **(a)** Many candidates struggled to provide accurate and concise explanations of these two terms. Web browser was better understood than the WWW.

(b) The majority of candidates, while familiar to some extent with client-side and server-side processing, found it difficult to provide detail or distinguish clearly between them using relevant DT terminology.

(c) Most candidates were familiar with CSS but a common weakness was a failure to address the specific context of the question – the use of CSS to ensure that the appearance of all pages in a website is consistent. As a result, few candidates achieved top marks.

(d) The use of private and public keys for encryption of messages was understood better than in previous years and some excellent responses were noted. However, the candidature was evenly split, with an equal number of candidates failing to distinguish between the two types of key.

Assessment Unit A21 Information Systems

Unit Overview

These comments reflect the opening comments about the extremely difficult circumstances faced by teachers and candidates as they prepared for this set of examinations.

Q1 **(a)** The majority of candidates were familiar with the three types of network. However, many candidates failed to distinguish clearly between them as was required. Phrases such as 'bigger than' or 'smaller than' are not appropriate at this level.

(b) The overall level of response to this question was very high. The format of the question was designed to reduce the amount of writing required of candidates and in this it appeared to be successful.

(c) The majority of candidates could describe the two types of network and some excellent responses were noted.

(d) This QWC question required candidates to evaluate the star and ring network topologies with respect to the failure of a single cable. The question required a concise, focused answer using the relevant DT terminology. As was expected, only a very small number of candidates met the very strict criteria for a top-band response. Also, as was intended, only a few more met the strict criteria for a middle-band response.

Q2 **(a)** A very high proportion of candidates was unfamiliar with the responsibilities of the ITU, and much guessing was apparent.

(b) The overall level of response to this question was very high.

(c) In general, candidates were familiar with the structure of the OSI model and many correct responses were noted. The format of the question was designed to reduce the amount of writing required of candidates and in this it appeared to be successful.

(d) Candidates were generally familiar with these two layers of the OSI, but many failed to provide the required detail with the result that few complete responses were noted.

(e) A very high proportion of candidates was unfamiliar with VoIP with the result that only a very few candidates achieved any marks.

(f) This QWC question required candidates to compare the use of radio frequency identification (RFID) and barcodes in monitoring the location and levels of stock in a warehouse. The question required a concise, focused answer using the relevant DT terminology. As was expected, only a very small number of candidates met the very strict criteria for a top-band response. Also, as was intended, only a few more met the strict criteria for a middle-band response.

Q3 **(a)** A high proportion of candidates could explain why the data in the table was not in 1NF but many of these failed to use appropriate DT terminology.

(b) Primary key was better understood than composite key although many candidates struggled with the appropriate DT terminology.

(c) Many candidates could explain why the data in the table was not in 2NF but some of these failed to use appropriate DT terminology.

(d) Many candidates could explain why the data in the table was not in 3NF but some of these failed to use appropriate DT terminology.

(e) The overall level of response to this question was high and many complete responses were noted. It must be stressed, however, that normalized entities must be named, and have their key field identified along with all other attributes.

(f) A high proportion of candidates struggled to explain a foreign key using appropriate DT terminology.

(g) The standard of ER model produced in this question was very high and many complete solutions were noted.

(h) This QWC question required candidates to compare the use of SQL and QBE for creating a database and generating queries. The question required a concise, focused answer using the relevant DT terminology. As was expected, only a very small number of candidates met the very strict criteria for a top-band response. Also, as was intended, only a few more met the strict criteria for a middle-band response.

Q4 **(a)** Many candidates struggle to state the purpose of the Turing Test concisely and accurately.

(b) The majority of candidates could explain how the Turing Test is carried out and many correct and complete responses were noted.

(c) In general, candidates were familiar with these components of an expert system, but many struggled to provide the required detail.

(d) There was confusion about the role of insurance consultants in the development of the expert system. Many candidates seemed to think the consultants would design the system, or be primarily involved in testing the system.

(e) This QWC question required candidates to evaluate the use of an expert system when calculating insurance premiums. The question required a concise, focused answer using the relevant DT terminology. As was expected, only a very small number of candidates met the very strict criteria for a top-band response. Also, as was intended, only a few more met the strict criteria for a middle-band response.

Q5 **(a)** In general, candidates were not appropriately familiar with the purpose of natural language processing.

(b) In general, candidates could not describe two appropriate uses of natural language processing.

(c) Many candidates made the mistake, which has been highlighted in previous Chief Examiner Reports, of confusing robotics with robots with the result that only a small number of responses attracting more than half the available marks was noted.

(d) This QWC question required candidates to evaluate the use of robotics in exploring another planet. The question required a concise, focused answer using the relevant DT terminology. As was expected, only a very small number of candidates met the very strict criteria for a top-band response. Also, as was intended, only a few more met the strict criteria for a middle-band response.

(e) Only a small number of candidates could describe the communications technology enabling an uninterrupted phone call to be made between two mobile phones with the result that very few accurate and complete responses were noted.

Q6 **(a)** The overall level of response to this question was very good.

(b) A high proportion of candidates struggled to identify two different ways in which the retail sector uses big data.

(c) The overall level of response to this question was disappointing. Few candidates focused on the threat to the privacy of individual shoppers with the result that their responses were really about DP legislation.

(d) Most candidates were familiar with cloud computing but many candidates failed to focus on its use by organisations rather by individuals.

(e) Most candidates were not familiar with the terms virtualisation and hosted instance. Few candidates gained any marks in this question.

(f) Many candidates were familiar with the main ethical problems which could arise from automated decision making and some excellent responses were noted.

Principal Moderator's Report

Assessment Unit A2 2 Application Development

Overview

This module was reduced in content for the Summer 2022 series. The rationale behind the adaptation was due to the COVID-19 pandemic, as it was recognised that both Candidates and Centres have been impacted by the pandemic. The adaptation made this module more accessible to candidates across the full grade range and also reduced the amount of documentation required in the solutions. There was still a requirement to provide evidence using a high level of problem-solving skills as expected at A2. The adaptation also retained a reduction in teacher assessment at each stage of the A2 solutions.

The following content was omitted from 2022 A2 Case Study and the remaining content made up 50% of the original module:

- Produce a Gantt chart to schedule the project.
- Prioritise and justify user requirements into essential and non-essential.
- Specify the hardware and software requirements to produce the new system.
- Produce an overall design for the system.
- Produce a description for each process, and algorithms where appropriate, based on user requirements.
- Design the required queries and reports.
- Document a strategy for system implementation.
- Identify possible extensions to the system.
- Develop an electronic guide, accessible from the user interface, that explains how to use the application.

Candidates carried out a problem-solving exercise based on a pre-released Case Study. The case study used this year provided different pathways for Candidates to demonstrate their knowledge and skills in problem solving. The evidence of the solutions indicated that it worked as intended in that Candidates were able to focus on a particular pathway such as a member booking system or product ordering and invoicing system. There was also evidence in some Centres of using a "template" based solution meaning Candidates followed the same steps in creating a solution. This practice should be discouraged as the case study is designed to allow Candidates to create their own individual solution with teacher guidance.

In terms of the application of the reduced assessment criteria most Centres applied the criteria accurately and awarded marks within the agreed standards set by the Moderation team. Centres should note the reporting back of marking on the TAC 6 report is also useful in preparation for assessing the next cohort. Centres where marks were adjusted should be encouraged to attend Agreement Trials, use the guidance on CCEA website and submit sample work to the Portfolio Clinic in February/March 2023. The use of this clinic is proving to be a positive experience for Centres in ensuring that the application of the assessment criteria is accurate in the summer series. Teachers should also be encouraged to add comments in the textboxes provided justifying the mark awarded to each assessment criteria. The Moderation team found this very useful in helping to make judgements.

Administration continues to be of a high standard in many Centres with Candidate solutions clearly labelled in electronic folders referenced by Candidate number identifying the top and bottom candidates on their electronic submission. Centres should continue to submit two documents in each Candidate folder - one should be the documentation whilst the other should be the database solution.

Analysis

It was pleasing to see Candidates clearly identifying internal and external constraints for their solutions. Candidates who were awarded higher marks in this section were also able to detail how the constraints described may impact on their solutions. Clearly identified constraints allowed Candidates to produce a more detailed set of User Requirements in terms of outputs expected, inputs needed, and a clear explanation of the processes required. There was also evidence of the inclusion of very detailed and easy to follow DFDs, both at Level 0 and Level 1 to illustrate the flow of data through the planned solution and the clear identification of the processes required.

Design

The majority of Candidates produced a detailed Data Dictionaries for each entity including a clear reference to data validation checking for attributes. High level solutions included around six tables as evidenced in the normalisation of data and detailed in the fully labelled ER diagrams for the relational database system. It was pleasing to see solutions that referenced and justify each stage of the normalisation process with clearly labelled primary and composite keys. Candidates who had detailed their User Requirements in the Analysis stage were able to develop a comprehensive test plan and go on to use integration testing to test the systems functionality against user requirements.

Application Development

It was pleasing to see evidence of excellent relational database solutions produced by many Candidates. All Centres used ACCESS as their database software to develop and implement their solutions. Many Candidates had included an intuitive interface which assisted the Moderation team as they navigated through the solutions.

Documentation and Evaluation

Evaluations were generally of high quality including detailed descriptions of what the solution achieved, together with evaluating each of the user requirements. Some candidates were able to refer to detail benefits and areas for improvement for their solutions.

Contact details

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